BST Construction from Given Elements

Output Analysis

Time Complexity

• BST construction by repeated insertion: $O(n^2)$ in the worst case; O(n log n) on average for a reasonably balanced insertion order.

• Displaying the tree after every insertion: $O(n^2)$ total if each display visits all nodes after each insertion.

• Inorder traversal: O(n).

• Preorder traversal: O(n).

• Postorder traversal: O(n).

• Height calculation, leaf counting, and searching: O(n), O(n), and O(h) respectively; search is O(n) in the worst case.

• Deletion: O(h), where h is the height of the BST; O(n) in the worst case.

Space Complexity

• BST storage: O(n).

• Recursive insertion, traversal, height, counting, search, and deletion use O(h) call-stack space.

• Overall auxiliary space: O(h), excluding the memory used to store the tree.

Here, n is the number of nodes and h is the height of the BST.